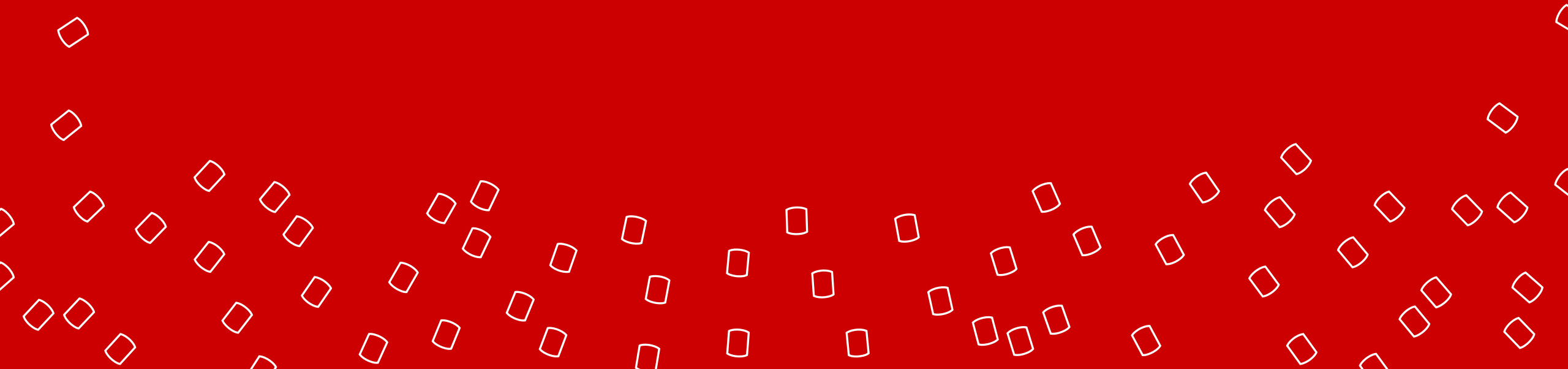


Fun With Radios: PostgreSQL, PostGIS and AWS Aurora



Grant Fritchey

DevOps Advocate

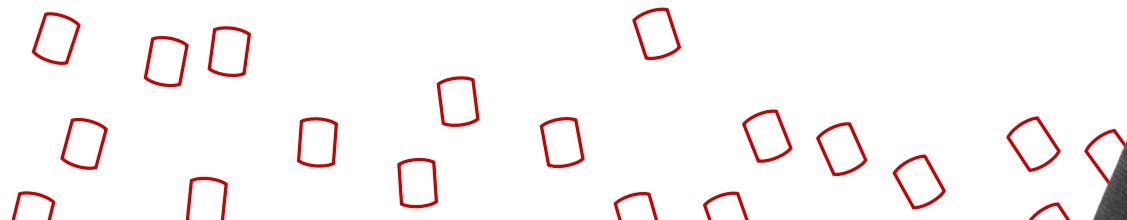
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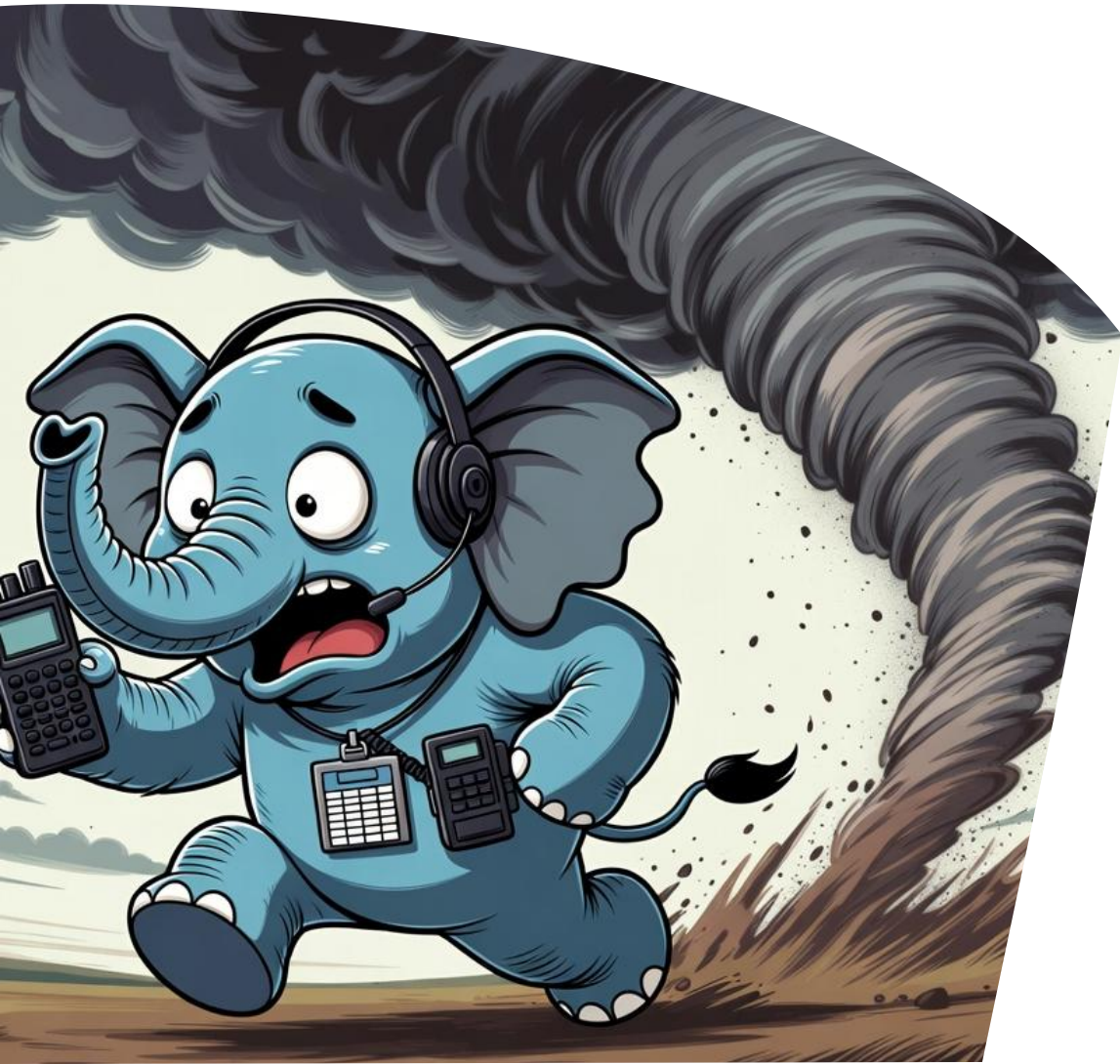


| Session Goals



- Learn about the Automatic Packet Report System (APRS)
- Understand how AWS Aurora can be used in combination with PostgreSQL
- See how the postGIS extension works with geographical data

| Why Radios



- Oklahoma (aka Tornado Alley)
- SKYWARN
- Emergency Comms

| OK, But Really, Why Radios?



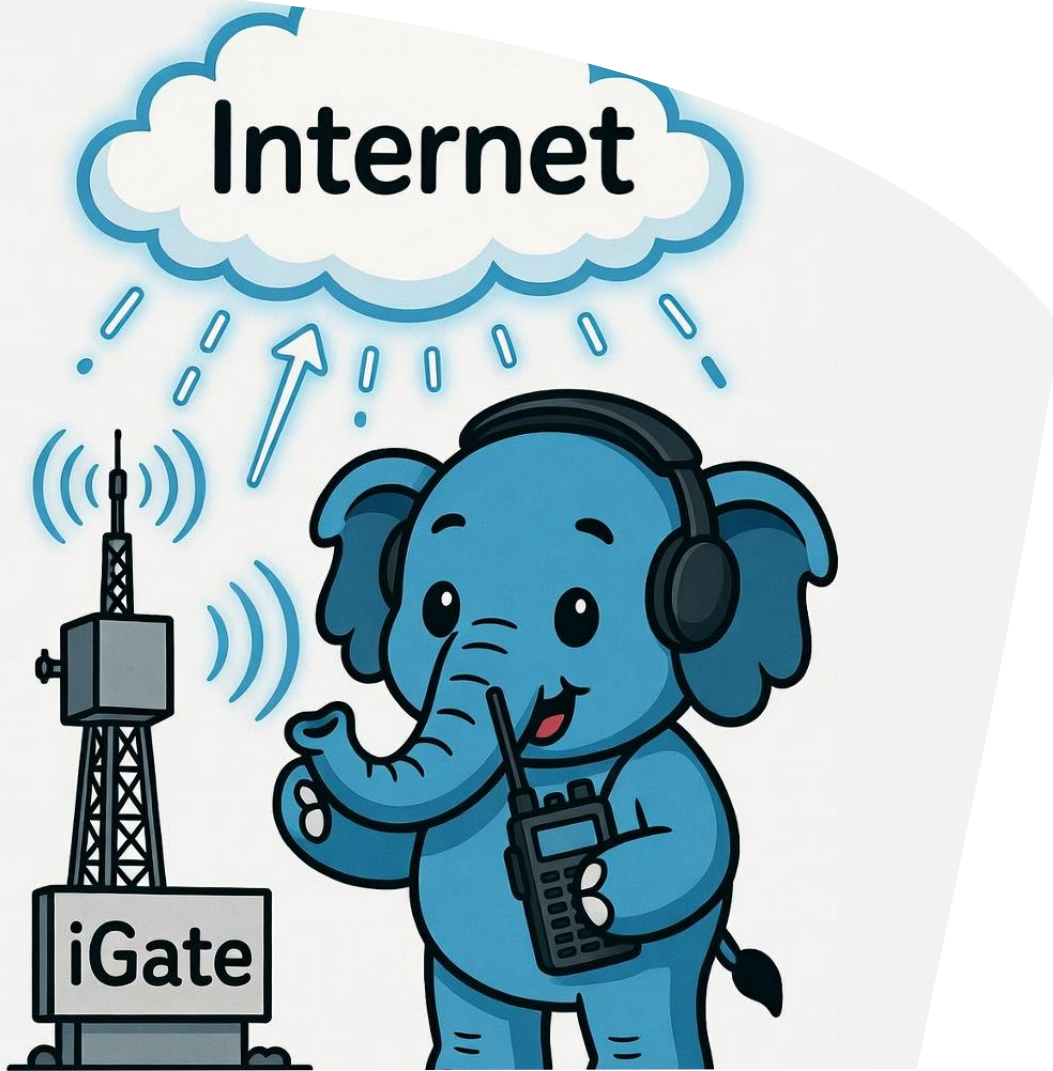
- First, I'm a giant nerd
- Second, I wanted an excuse to play with radios at work

| Automatic Packet Reporting System



- APRS
- A digital transmission sent over analog signals
- Short messages
- Weather reports
- Location data

| How APRS Works



- Terminal Node Controller (TNC)
- Packets
 - AX.25
 - 1200 baud
- Internet Gateway (iGate)
 - One way (receive only)
 - Two way (transceiver)
- APRS-IS

| Common Hardware



- APRS capable radios
- Radio + TNC + phone/computer
- Dedicated hardware trackers
- Home weather stations
- RTL-SDR Software Defined Radio

Demo

Bridgecom
Maverick




APRS
Transmission
iGate




APRS
Network



| My App

- AWS Aurora
- PostGIS
- Lambda Functions
 - And the rest
- Agentic AI



| AWS Aurora

- PostgreSQL
- Split storage & compute
- Readable secondaries
- High performance
- Connection management
- Limited on Extensions



| PostGIS

- Geography w/ spherical calculations
- GIST indexing
- Distance & speed calculations
- GeoJSON



| Lambda Functions

- Serverless
- Focused
- Easy API call



| And the rest

- Security
 - Secrets Manager
- S3
 - Nominatum
 - Leaflet
- Automation
 - EventBridge



| Agentic AI

- Automation
- Filling gaps
- Keep a human in the loop



| DEMO



| Wrap-up

- Radios are fun
- AWS Aurora is mostly just PostgreSQL
- Agentic AI is wonderful, but not trustworthy



Code & Slides



| Session Goals



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